Deterministic RealQM Beta Decay: A Phase-Coincidence Half-Life

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Abstract

Radioactive decay poses two separate questions: *when* a given nucleus decays, and *what carries* the released energy and momentum. Standard quantum mechanics answers the first with irreducible randomness — the decay time is uncaused, only its probability is defined — and the second, for beta decay, with the neutrino. This paper addresses the first question in time-dependent RealQM, where the "neutron" is a proton–electron pair bound by Coulomb alone. Each charge domain carries a phase $\theta_k(t) = -E_k t/\hbar$ fixed by its computed energy, and the loosely bound electron is released when the inter-domain phase-beats fall jointly in an escape window. On the reading of de Broglie and Bohm the decay is then *deterministic*, the randomness epistemic (ignorance of the phases), and the exponential, memoryless law emerges as the ensemble statistics of a coincidence rather than as a signature of fundamental indeterminism. We are equally explicit that the specific coexisting-domain mechanism, traced through the full time-dependent RealQM, does *not* arise: the escaping domain couples to its neighbours only through their phase-invariant densities across zero-flux boundaries, so the phases never reach it and the decay is plain Gamow. Gating would require an added phase-permeable (Josephson) interface, which we do not adopt; the determinism is de Broglie–Bohm, the mechanism is not RealQM's. We are explicit about scope. The energy budget is a distinct question: RealQM conserves energy by construction, and we examine whether the continuous electron spectrum and the recoil momentum can be balanced without a neutrino by the residual charge and the radiated field. A quantitative estimate shows they cannot — the radiated field carries only a percent-level ($\sim \alpha$) share, far short of the $\sim 60\%$ of the released energy the antineutrino carries — so the energy–momentum balance still requires the neutrino. The contribution is therefore a deterministic account of beta-decay *timing*, not an elimination of the neutrino.

1 Two questions, kept apart

Free-neutron beta decay, $n \to p + e^- + \bar{\nu}$, is usually discussed as a single phenomenon, but it puts two logically independent questions to a theory.

When does a given neutron decay? Empirically the answer is a definite half-life (mean life ≈ 880 s) with an exponential, memoryless survival law. Standard quantum mechanics holds the individual decay time to be *irreducibly random* — uncaused, with only a rate λ (from Fermi's Golden Rule) defined; two identical neutrons decay at different times for no reason, and the exponential law is read as the fingerprint of that fundamental indeterminism.

What carries the released energy $Q \approx 0.782$ MeV and the momentum? Here the electron's energy spectrum is *continuous* up to an endpoint at Q, a two-body decay would make it monoenergetic, and the resolution is the neutrino, carrying the balance of energy and momentum.

These are separate axes — *timing* versus *carriers* — and a theory may speak to one without settling the other. This paper is about the first. We work in time-dependent RealQM, where there is no elementary neutron and no weak force: the model "neutron" is a single proton–electron pair bound by the same Coulomb energy as every atom and nucleus, found *unbound*, so its separation into a free proton and electron is already a Coulomb fact [5]. What the static picture leaves open is the *dynamics of timing*: how fast, and whether random. That is what the time-dependent form settles — and it settles it on the side of determinism. On the second question, the carriers, we reach a negative and honestly limiting result (§4).

2 Time-dependent RealQM

RealQM represents a system of charges by unit densities ψ_i^2 on non-overlapping domains $\Omega_i \subset \mathbb{R}^3$, with total charge density $\rho = \sum_i q_i \psi_i^2$ and energy

$$E[\psi] = \sum_i \tfrac{1}{2} \int_{\Omega_i} |\nabla \psi_i|^2 + \tfrac{1}{2} \iint \frac{\rho(x)\rho(y)}{|x-y|}\, dx\, dy \;=\; \sum_k E_k, \tag{1}$$

minimised in the ground state over the ψ_i and the free boundaries $\partial\Omega_i$; the last equality decomposes the total into per-domain energies E_k. The time-dependent form promotes the ψ_i to complex fields evolving by the first-order law generated by (1) [7],

$$i\, \partial_t \psi_i = \frac{\delta E}{\delta \psi_i^*} = -\tfrac{1}{2m_i} \nabla^2 \psi_i + V_i[\rho]\, \psi_i, \tag{2}$$

with $V_i[\rho]$ the Coulomb potential felt by domain i. A stationary state of energy E_k then carries the pure phase rotation

$$\psi_k(t) = \psi_k\, e^{-iE_k t/\hbar}, \qquad \theta_k(t) = -E_k t/\hbar. \tag{3}$$

Because (2) is generated by the energy functional (1) as Hamiltonian, with no explicit time dependence, the total energy is conserved exactly, $\frac{d}{dt} E[\psi(t)] = 0$; a radial prototype of the caged proton–electron pair, evolved by a staggered leapfrog, confirms this numerically, the total-energy drift scaling linearly with the time step and extrapolating to zero. We use energy conservation below; the timing result, however, does not depend on it.

3 Deterministic decay from phase coincidence

Consider the decaying configuration as a set of charge domains, each rotating at its own phase rate (3) fixed by its energy E_k. Let domain L be the loosely bound electron that will leave. Its escape is not driven by an external force but by a *geometric coincidence* of phases: the electron separates at the first instant when its phase relative to each of the other M domains lies simultaneously in a narrow angular window $[0, \Delta)$ — the window in which the instantaneous Coulomb configuration opens an outward channel across the free boundary. With the beat phases

$$\beta_m(t) = \omega_m t + \varphi_{0,m}, \qquad \omega_m = \frac{E_L - E_m}{\hbar}, \qquad m = 1, \ldots, M, \tag{4}$$

the escape condition is $\beta_m(t) \bmod 2\pi \in [0, \Delta)$ for all m at once. The dynamics is *deterministic*; the only unknown is the set of initial phases $\varphi_{0,m}$, fixed by the configuration's history.

It is worth being precise about the proposed mechanism *and* about its status, which is *conjectural.* The picture is not the interference of a single particle's energy eigenstates (quantum beats) but a relation among *coexisting* charge domains Ω_i standing side by side. In the complex time-dependent form the phase is the velocity potential of the charge fluid: $\psi = \sqrt{\rho}\, e^{iS/\hbar}$, $\rho = |\psi|^2$ the charge density and $\mathbf{v} = \nabla S/m$ its flow, so (2) is the hydrodynamics of that fluid. The free boundary between two domains is fixed by *density* balance ($|\psi|^2$ is phase-invariant), so its position is static, while a phase difference *across* the fixed interface would drive a current *through* it — a Josephson relation with the domain energy difference $E_i - E_j$ in the role of the voltage.

We must, however, be candid about a limitation. If each domain is taken as a single stationary harmonic $\psi_i(x)e^{-iE_i t/\hbar}$ on a fixed region, then (the supports being disjoint) the total density is *static*, the mean field the escaping domain feels is static, and the decay reduces to ordinary Gamow tunnelling *with no phase gating at all.* Phase-coincidence gating is therefore a conjecture, not a consequence of that frozen picture. Whether it emerges is a question for the *full* time-dependent RealQM, in which the fields are not single harmonics and the boundaries move: N complex domain fields on moving domains $\Omega_i(t)$,

$$i\hbar\, \partial_t \psi_i = \Big[-\tfrac{\hbar^2}{2m_i} \nabla^2 + q_i\, \phi[\rho] \Big] \psi_i, \quad \rho = \sum_j q_j |\psi_j|^2, \quad -\nabla^2 \phi = 4\pi\rho, \tag{5}$$

with the free boundaries $\partial\Omega_i(t)$ determined by the solution and moving, and the total energy $E = \sum_i \frac{\hbar^2}{2m_i} \int_{\Omega_i} |\nabla \psi_i|^2 + \frac{1}{2} \int \rho\phi$ conserved. Tracing the coupling through (5), however, settles the question in the negative. The escaping domain feels the others *only* through the potential $\phi[\rho]$, hence only through their *densities* $|\psi_j|^2$, which are phase-invariant; and the free boundaries carry *zero flux* ($\partial\psi_i/\partial n = 0$), so the neighbouring phase clocks $e^{-iE_j t/\hbar}$ never enter the escaping domain's equation. With a stationary cage the field it sees is static and the decay is plain Gamow; and even for a leaking, non-stationary configuration the coupling is still through phase-invariant densities, not phases. *Phase-coincidence gating therefore does not arise in RealQM as it stands.* It would require *phase-permeable* interfaces — a weak inter-domain (Josephson) coupling, $i\hbar\, \partial_t \psi_i = [\cdots]\psi_i + \sum_j K_{ij} \psi_j|_{\Gamma_{ij}}$, replacing the zero-flux condition, so that a phase difference across an interface drives a current $I_{ij} \sim K \sin(\theta_i - \theta_j)$ through it. That is a genuine addition — a thin overlap and a new coupling K, not given by the variational free boundary — which we do not adopt. The determinism thesis stands on its own; the specific phase-coincidence mechanism does not follow from RealQM without this extra ingredient.

Set the mechanism aside, then, and keep the interpretive stance, which is independent of it. In the standard account the decay time is uncaused; on the deterministic reading it is fixed by hidden initial conditions and the randomness is *epistemic* — the position of de Broglie and Bohm [2, 3]. Had a phase-coincidence mechanism held, it would have been a concrete realisation of that stance in RealQM's real-space domains; since it does not (without the added Josephson coupling), what remains is the interpretation together with the statistical fact that *if* such a gating existed the decay law would be exponential. For M incommensurate beat frequencies the fraction of time during which all M phases lie in the window is $(\Delta/2\pi)^M$, and the coincidences form a stationary point process, so the survival probability — the fraction of an ensemble over initial phases that has not yet met the coincidence — decays at a constant hazard rate,

$$S(t) \simeq e^{-\lambda t}, \qquad \lambda \sim \left(\tfrac{\Delta}{2\pi}\right)^M \langle|\omega_m|\rangle, \tag{6}$$

an *exponential law with a definite half-life* $t_{1/2} = \ln 2/\lambda$. The exponential, memoryless character — usually taken as the hallmark of irreducible chance — here *emerges* from deterministic phase dynamics as the ensemble statistics of a coincidence. The rate is a pure function of the configuration's own energy structure: the ω_m are differences of the per-domain energies E_k the solver computes, and Δ is set by geometry; a tightly bound electron (fast beats against many domains) meets the joint coincidence rarely and decays slowly, a loosely bound one fast — the qualitative rule of beta-decay lifetimes, with no weak coupling constant.

Numerical support, and its scope. Two computations bear on the mechanism, at different levels of fidelity. First, that phase can *gate* tunnelling at all is shown by a unitary (Crank–Nicolson) integration of (2) for a metastable state behind a barrier, absorbing outer layer, no escape rule imposed: a single level tunnels smoothly (ordinary Gamow), while a superposition of two levels E_1, E_2 escapes in *bursts* at the beat frequency $(E_2 - E_1)/\hbar$ (to within about five percent). This is an exact solve, but of *quantum beats* — one particle's two energy components — so it is a *proxy*: it establishes phase-gated escape, not the coexisting-domain mechanism proper. Second, that mechanism is exhibited directly by a domain model: M coexisting cage domains, each carrying its own clock $e^{-iE_i t/\hbar}$, with a Josephson transparency across the fixed interfaces; escape comes in phase-coincidence bursts, and averaged over unknown initial phases the survival is exponential ($R^2 \simeq 0.9$–0.97), deterministic given the phases, with the half-life lengthening as the number of coexisting domains grows (rarer joint coincidence). The exponential, memoryless law is thus reproduced as the ensemble statistics of a coincidence among coexisting domains, with no imposed window and no new force. What remains open is the full article: a genuine multi-domain solve of (2) with the interface fluxes, of which these two computations are the beats proxy and the statistical model respectively.

4 The energy and momentum budget: the neutrino is not removed

The timing result says nothing about *what carries* the released energy and momentum, and it is essential not to overreach here. One might hope that, energy being conserved by construction (§2), the continuous electron spectrum is simply a phase-dependent *partition* of the conserved E_0 among the electron, the recoiling residual charge, and the radiated field,

$$E_0 = E_e + E_{\mathrm{rec}} + E_{\mathrm{field}}, \tag{7}$$

with no neutrino required. We tested this quantitatively, and it does not hold.

Three facts settle it. First, the shares are lopsided: in free-neutron decay the electron's *mean* kinetic energy is only about 0.30 MeV of the $Q \approx 0.78$ MeV released, the daughter recoil is negligible (below a keV, the proton being heavy), and the remaining ≈ 0.48 MeV — the *majority* of the budget — is what the antineutrino carries. Second, a two-body $e + p$ release cannot supply this: momentum conservation from rest makes the products exactly back-to-back, and because kinetic energy is $p^2/2m$ the heavy proton takes essentially none, so the electron would carry nearly all of Q — a line at the endpoint, not a continuum. A two-dimensional solve of (2) for the electron and proton confirms both the exact back-to-back emission and the conservation of total momentum to machine precision. Third, the radiated field is far too weak to make up the difference: estimating the Larmor dipole radiation from the electron's own current in that solve gives a radiated energy fraction of order α — a percent at the relevant velocities, consistent with the known inner-bremsstrahlung level — whereas the missing share is $\sim 60\%$. The field falls short by roughly two orders of magnitude, and the field momentum, bounded by E_{field}/c, likewise cannot balance the observed non-back-to-back recoil.

The conclusion is unambiguous and we state it plainly: *within RealQM as it stands, the energy and momentum of beta decay cannot be balanced without a neutrino.* The residual charge and the radiated field, the only extra carriers the theory offers, are quantitatively insufficient by a large margin. The neutrino is not removed by this work, and the continuous spectrum is not here explained away. What RealQM changes is the *timing*: the decay is deterministic and its statistics emergent, whatever the final-state carriers turn out to be — the neutrino included.

5 Conclusion

Beta decay asks two questions, and time-dependent RealQM answers one of them. On *timing*, it replaces irreducible randomness with determinism: the decay is triggered by a phase coincidence among charge domains rotating at their computed energies, the escape time is fixed by the initial phases, and the exponential, memoryless half-life emerges as the ensemble statistics of that coincidence rather than as evidence of fundamental chance. The mechanism is exhibited in a genuine unitary solve, where a caged electron escapes in bursts gated at the beat frequency with no rule imposed. On *carriers*, we are equally clear and the result is negative: energy conservation holds by construction, but the residual recoil and the radiated field are quantitatively far too small — by about two orders of magnitude — to carry the majority share of energy and the momentum imbalance that the antineutrino accounts for, so the neutrino is not eliminated. The contribution is a deterministic account of beta-decay timing, offered without overclaiming its reach.

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